

Nirbhay Beri

Senior Engineer — Digital Transformation & Industry 4.0 | Industrial AI & Smart Manufacturing | M.Tech, Smart Manufacturing, IISc Bangalore

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Professional Summary

Digital Transformation and Industry 4.0 professional with experience in smart manufacturing, Industrial AI, IIoT, MES integration, digital twins, and IT-OT convergence. Proven track record of developing connected manufacturing solutions that improve traceability, operational visibility, production efficiency, quality management, and data-driven decision making. Experienced in translating manufacturing challenges into scalable digital solutions using AI vision, edge computing, industrial communication protocols, cloud analytics, and smart factory technologies.

Professional Experience

Senior Engineer — Industry 4.0 & Smart Manufacturing

Jan 2026 – Present

Atlas Copco Group (GECIA), Pune

- Developed an **AI-assisted vision inspection and robotic rework validation workflow** for vacuum pump families using **NVIDIA Jetson Xavier**, industrial cameras, PLC-integrated pass/fail logic, and **UR Cobot integration**, enabling automated inspection handling for multiple product variants, reducing manual inspection dependency, and improving quality decision consistency.
- Designed a **Vision based Universal Kit Completeness Verification Solution** integrating industrial cameras, edge AI inference, operator dashboards, and traceability workflows, automating component verification and significantly reducing manual inspection effort for compressor sub-assemblies.
- Developed a real-time **BLE-based Asset Tracking System** for **1,000+ manufacturing trolleys**, enabling real-time visibility of inbound, outbound, production, and repair assets, reducing manual tracking effort by over 80%, improving asset utilization, and minimizing production delays associated with unavailable or misplaced trolleys.

Engineer — Industry 4.0 & Smart Manufacturing

Jul 2024 – Dec 2025

Atlas Copco Group (GECIA), Pune

- Led the conceptualization and development of the **GECIA Industry 4.0 Lab**, creating the technology roadmap, hardware architecture, vendor partnerships, and innovation framework for Industrial AI, IIoT, robotics, and smart manufacturing initiatives.
- Developing a Test Cell Data Analytics dashboard for AC Industrial Compressor test beds using Power BI, visualizing key quality and performance metrics such as SPC, FTR, and RFT (Ready for Testing) to support operator-driven testing and decision-making.
- Built an IIoT telemetry pipeline using **Azure IoT Hub** to ingest environmental data (temperature/humidity) from **Advantech WISE 4250** modules via MQTT; visualized real-time metrics in Power BI for operational monitoring.
- Design architecture of **MES-PLC communication** for AC New MES using **Kepware** drivers via **OPC UA**, enabling reliable WorkStart/Stop and Andon events; supported successful go-live across two **conveyor lines** at the PNE Plant.
- Created industrial process simulations using **Visual Components** and Python APIs for BMW, GM, and customer-center manufacturing scenarios, accelerating manufacturing layout validation and reducing engineering effort during production planning activities.
- Prototyped XR-based industrial visualization experiences using **Unity, WebSockets, and HoloLens 2**, enabling immersive product demonstrations and supporting future digital twin initiatives.

Research Intern — AI & Industrial Metaverse

May 2023 – Jul 2023

Siemens Technology, Bengaluru

- Developed a **virtual factory environment** in Unity integrated with HTC Vive to enable immersive visualization and evaluation of manufacturing layouts and production workflows.
- Built a machine-layout optimization framework using evolutionary algorithms to improve production efficiency, ergonomics, and safety considerations within factory planning environments.
- Contributed to industrial metaverse and digital twin initiatives focused on next-generation manufacturing visualization and smart factory simulation.

Selected Academic & Research Projects

Multimodal XR System for Remote Human-Robot Welding & Assembly

M.Tech Thesis

- Developed an XR-based human–robot interaction framework integrating visual, haptic, voice, and digital twin technologies for remote industrial robot operation.
- Built a VR-enabled welding and assembly environment synchronized with robotic systems using Unity, HoloLens 2, HTC Vive, and custom APIs.
- *Technologies: Unity, XR/VR, Digital Twin, HoloLens 2, HTC Vive, Python, Robot Teleoperation, Computer Vision.*

AI & Vision-Based Robot for Surface Defect Detection

IISc Bangalore

- Developed a robotic AI inspection system using computer vision and automated robot scanning for defect detection on aerospace components.
- Implemented YOLO-based defect classification, calibrated image acquisition pipelines, and robot path planning for automated industrial quality inspection.
- *Technologies: YOLOv8, OpenCV, Robot Vision, Camera Calibration, Python, Industrial Inspection.*

Technical Expertise

Automation & Industry 4.0

IT/OT Convergence, OPC UA, MQTT, Basic PLC Programming (TIA Portal, SIMATIC Manager, ISP Soft), Ignition, Visual Components (Simulation), Kepware, OPC Router.

AI, Robotics & Computer Vision

UR10 Cobot, DOBOT Magician, FANUC Robots, Machine Learning (TensorFlow, Scikit-Learn), Computer Vision, NLP, GenAI, Data Analysis (Pandas, NumPy, Matplotlib).

Cloud, Software & Platforms

Microsoft Azure (IoT Hub, Logic Apps, Functions, Stream Analytics, ML), Unity 3D, Node-RED, Power Platform, SQL Server, MongoDB, InfluxDB, Grafana.

Hardware & Edge Computing

HoloLens 2, HTC Vive, Jetson Xavier/Nano, IoT Modules (WISE 4250), Industrial Cameras, Haptic Devices (3D Systems Touch).

Programming & Methodologies

Python, C#, GitHub, Cross-functional Collaboration, Innovative Problem Solving.

Key Academic Coursework

Digital Manufacturing, Sensors & Transducers, HMI, Operations Management, Additive Manufacturing, Creative Engineering Design, Probability & Statistics, Computing for AI & ML, NLP, Computer Vision, Data Modelling.

Education

M.Tech — Smart Manufacturing, Indian Institute of Science (IISc), Bengaluru **2022 – 2024**

With Distinction, CGPA: 8.7/10

GATE 2022 – All India Rank 26

B.Tech (Honors), Dr. A.P.J. Abdul Kalam Technical University (AKTU), Lucknow **2018 – 2022**

CGPA: 9.0/10

Publications

- **HaM3D: Generalized XR-Based Multimodal HRI Framework** — Journal on Multimodal User Interfaces, Springer, 2024.
- **Enhancing Production Efficiency Using GA-Assisted Visualization** — I-4AM Conference, Springer, 2023.

Achievements & Certifications

- **Star of the Month**, Atlas Copco Group, Aug 2025.
- **Siemens Technology Research Fellowship**, 2022–2024.
- **GATE 2022 — All India Rank 26.**
- **B.Tech Honors** — awarded Honors Degree among top-performing students.
- **NPTEL Industrial IoT 4.0** — Top Rank.